# INFOGRAPHIC: Systematic Reviews & Meta-Analysis

*The Pinnacle of Evidence-Based Statistical Modelling*

**[ SYSTEMATIC REVIEW ]**
A qualitative, structured framework to identify, critically appraise, and synthesize all relevant global studies on a specific question.

**[ META-ANALYSIS ]**
The quantitative, statistical core that pools data from those independent studies to calculate a single, high-power pooled effect size.

## 1. The PRISMA Flowchart Pipeline

Every legitimate systematic review must filter global research using a rigid, reproducible screening protocol:

1. **Identification:** Searching massive databases (PubMed, Scopus) using strict keywords.
2. **Screening:** Removing duplicates and checking titles/abstracts against inclusion criteria.
3. **Eligibility:** Reading full-text articles to filter out poor methodology or irrelevant setups.
4. **Included:** The final subset of high-quality papers selected for qualitative review and statistical pooling.

## 2. Statistical Modelling: Fixed vs. Random Effects

When mathematically pooling different studies together, we must choose how we model the underlying variances.

| Fixed-Effect Model | Random-Effects Model |
|---|---|
| Assumes all studies are trying to measure the **exact same true effect size**. | Assumes the true effect size **varies between studies** due to population, dosage, or geographic differences. |
| Source of variation: Only within-study sampling error ($\varepsilon_i$). | Source of variation: Sampling error ($\varepsilon_i$) + between-study heterogeneity ($\tau^2$). |
| Larger studies heavily dominate the weights. | Weights are balanced out, giving small studies a fairer voice. |

## 3. Deciphering the Forest Plot

The Forest Plot is the ultimate visual output of a Meta-Analysis. It reads like a graph of confidence intervals stacked vertically.

- **Individual Horizontal Lines:** Represent each individual study's calculated effect size (square box) and its 95% Confidence Interval (whiskers).
- **Box Size:** Proportional to the study's statistical weight (larger sample = bigger box).
- **The Vertical Line of No Effect:** Marked at 0 for differences (e.g., mean differences) or 1 for ratios (e.g., Odds Ratio, Relative Risk). If a study's line crosses this vertical axis, it is not statistically significant.
- **The Diamond at the Bottom:** The holy grail! The center of the diamond represents the combined pooled effect size, and its width represents the pooled 95% confidence interval.

## 4. Quantifying Heterogeneity ($I^2$)

Before trusting a pooled result, we must measure how much the studies disagree with one another mathematically.

> **The $I^2$ Statistic:** Measures the percentage of total variation across studies that is due to true heterogeneity rather than pure chance/sampling error.
>
> - $I^2 \leq 25\%$: Low heterogeneity (Fixed-effect model usually safe).
> - $I^2 \sim 50\%$: Moderate heterogeneity.
> - $I^2 \geq 75\%$: High heterogeneity (Random-effects model mandatory; pooling might even be questionable).

## 5. Publication Bias & The Funnel Plot

Studies with negative or neutral results are often locked away in drawers and never published. This distorts meta-analyses.

- **The Funnel Plot:** Plots study precision (sample size) on the Y-axis against the effect size on the X-axis.
- **Symmetry Check:** High-precision studies cluster tightly at the top. Low-precision studies scatter widely at the bottom, forming an inverted funnel. If one bottom corner of the funnel is completely empty, it indicates **publication bias**.